■■ CHEMICAL REACTIONS

Endothermic vs Exothermic

Detailed Lesson Plan — Grade 4 | Fun Science Series

■ **Subject**	Science — Chemistry (Grade 4)
■■ **Duration**	45–60 minutes
■ **Learning Goals**	Classify reactions; identify heat flow
■ **Materials**	Baking soda, vinegar, thermometer, cups
■ **Assessment**	Sorting activity + exit ticket quiz

■ SECTION 1: LEARNING OBJECTIVES

1. Define what a chemical reaction is and give everyday examples.
2. Explain the difference between exothermic and endothermic reactions.
3. Use memory hooks (Exo=Exit, Endo=Enter) to classify reactions quickly.
4. Identify signal words (gets hot, absorbs heat, feels cold) in descriptions.
5. Correctly sort at least 4 out of 6 example reactions into the right category.
6. Make real-world connections to daily Indian life and cultural contexts.

■ SECTION 2: BACKGROUND KNOWLEDGE FOR TEACHERS

What Is a Chemical Reaction?

A chemical reaction is a process in which one or more substances (called **reactants**) are converted into one or more new substances (called **products**). The key indicator is that a new substance is formed — with different properties from the starting materials. This is represented as:

Reactants → Products

Common misconception to address: Physical changes like melting ice or dissolving sugar are NOT chemical reactions because no new substance is formed. The original substance can be recovered.

Energy and Chemical Reactions

All chemical reactions involve an exchange of energy with the surroundings, usually as heat. Whether the surroundings feel warmer or cooler depends on the direction of that energy flow:

■ Exothermic Reactions	■ Endothermic Reactions
Heat is **released** to surroundings	Heat is **absorbed** from surroundings
Surroundings feel **HOT**	Surroundings feel **COLD**
Energy of products < Energy of reactants	Energy of products > Energy of reactants
Examples: combustion, rusting, respiration	Examples: photosynthesis, evaporation reactions

■■ SECTION 3: DETAILED LESSON PLAN

■■ Phase 1 — Warm-Up / Hook — 5–7 min

• Begin with a relatable question: *"Why does your hand feel warm when you rub it fast? Why does a cold pack feel cold?"*

• Ask students to think of one thing that gets HOT and one thing that gets COLD in daily life.

• Record responses on the board under two columns: HOT and COLD.

• Tell students: *"Today we're going to solve the mystery of why some reactions make things hot and others make things cold!"*

■■ Phase 2 — Direct Instruction: Chemical Reactions — 8–10 min

• Define: *A chemical reaction happens when substances change into something new.*

• Write on board: **Reactants** → **Products**.

• Give Indian context examples: Roti cooking on tawa, burning diya, rusting iron gate, curd forming from milk, baking a cake.

• Emphasize the key rule: **A new substance must form.** Melting ice = NOT a reaction (it's still water). Dissolving sugar = NOT a reaction.

• Quick check: Ask 2–3 students to give their own example and say if it's a chemical reaction.

■■ Phase 3 — Exothermic Reactions 8 min

• Introduce term: **Exothermic**. Write on board: **Exo = Exit** → **Heat Exits** → **Gets HOT ■**

• Signal words to recognise exothermic reactions: *gets hot, burning, flame, releases heat, gives off heat.*

• Real-life examples from the lesson: Burning coal in sigri (keeps us warm in winter), Rusting iron gate (releases heat slowly), Cooking dal on gas stove.

• Ask students: *"Can you think of something at home that gets HOT when a change happens?"*

• Optional demo: Hold hands together and rub vigorously — heat is produced (exothermic friction effect).

■■ Phase 4 — Endothermic Reactions 8 min

• Introduce term: **Endothermic**. Write on board: **Endo = Enter** → **Heat Enters** → **Gets COLD ■**

• Signal words to recognise endothermic reactions: *gets cold, absorbs heat, cools down, needs heat input.*

• Real-life examples: Clay pot (matka) cooling water, baking soda + vinegar (glass feels cold), photosynthesis — plants absorb sunlight.

• Ask students: *"Why does water stay cool in a clay pot? What does the pot do with heat?"*

• Connect to home: Ice packs used on injuries — they absorb heat from the body to feel cold.

■■ Phase 5 — Sorting Activity 10 min

• Present the 6 sorting challenges from the lesson (also listed in Section 5 below).

• Students can work in pairs. Give each pair a sorting card or write in their notebooks.

• Circulate and guide students who are unsure — prompt with: *"Does it get hot or cold?"*

• Review answers together as a class, explaining reasoning for each.

■■ Phase 6 — Recap & Exit Ticket 7–10 min

• Do a quick class recap: *Chemical reaction = new substance; Exo = Exit = HOT; Endo = Enter = COLD.*

• Hand out the exit ticket (see Section 6) — 3 quick questions to check understanding.

• Assign Science Spy Challenge homework: Find ONE hot or cold reaction at home and record it.

• Preview next lesson: *"Next time, we'll do REAL experiments — get ready!"*

■ SECTION 4: KEY CONCEPTS & MEMORY HOOKS

■ EXOTHERMIC

EXO = EXIT

Heat EXITS the reaction

Surroundings get HOT

Signal words:

gets hot • burning • flame • releases heat • gives off heat

■ ENDOTHERMIC

ENDO = ENTER

Heat ENTERS the reaction

Surroundings get COLD

Signal words:

gets cold • absorbs heat • cools down • needs heat

Comprehensive Comparison Table

Feature	■ Exothermic	■ Endothermic
Heat Flow	Released (OUT) to surroundings	Absorbed (IN) from surroundings
Temperature Effect	Surroundings get WARMER	Surroundings get COOLER
Prefix Meaning	Exo = Exit	Endo = Enter
Energy of Products	Lower than reactants	Higher than reactants
Indian Life Example 1	Burning diya ■	Clay pot (matka) cooling water ■
Indian Life Example 2	Coal burning in sigri	Plants doing photosynthesis ■
Kitchen Example	Cooking dal on gas stove ■	Baking soda + vinegar (cold glass)
Signal Words	Hot, burning, flame, releases	Cold, absorbs, cools, needs heat

✏■ SECTION 5: CLASSROOM SORTING ACTIVITY

Instruct students to classify each reaction below as either **Exothermic** ■ or **Endothermic** ■. Ask them to underline the signal word(s) that helped them decide.

#	Reaction Description	Answer	Reason / Signal Word
1	Burning Diwali crackers — produces light and heat	**EXOTHERMIC** ■	Signal: produces heat, burning

2	Cold pack placed on a sprained ankle — pack feels cold	**ENDOTHERMIC** ■	Signal: feels cold, absorbs heat from body
3	Wood burning in Lohri bonfire — gives off warmth	**EXOTHERMIC** ■	Signal: gives off warmth, burning, flame
4	Plants making food using sunlight (photosynthesis)	**ENDOTHERMIC** ■	Signal: absorbs sunlight (energy input needed)
5	Cooking dal on the gas stove — pot gets very hot	**EXOTHERMIC** ■	Signal: gets hot, burning gas
6	Baking soda + vinegar mixed in a glass — glass feels cold	**ENDOTHERMIC** ■	Signal: feels cold, absorbs heat during reaction
7 ★	Rusting of an old iron gate left outside in rain	**EXOTHERMIC** ■	Releases heat slowly; a slow exothermic reaction
8 ★	Curd (yoghurt) forming from warm milk with starter	**EXOTHERMIC** ■	Fermentation releases heat; milk gets slightly warm

★ Bonus questions for advanced students.

■ SECTION 6: EXIT TICKET ASSESSMENT

Distribute at the end of class. Students answer individually in 5 minutes. Use responses to identify students who need more support.

Name: ________________________________ **Date:** _______________ **Class:** _______

Q1. What is a chemical reaction? Write one sentence.

Answer: ____

____

Q2. Circle the correct answer:

An exothermic reaction: a) Absorbs heat b) Releases heat c) Does not involve heat

An endothermic reaction: a) Gets hot b) Gets cold c) Stays the same

Q3. Classify each reaction — write EXO or ENDO:

a) Burning a candle: ____________ b) Ice pack on knee: ____________

c) Cooking roti on tawa: ____________ d) Mixing baking soda + vinegar (cold glass): ____________

Q4. Complete the memory hooks:

Exo = ________________ = Heat ________________ = Gets ________________

Endo = ________________ = Heat ________________ = Gets ________________

Q5. (Bonus) Give one example of an exothermic reaction from your daily life that was NOT mentioned in class.

Answer: ____

Exit Ticket — Answer Key (Teacher Copy)

Q	Expected Answer
Q1	A chemical reaction is when substances change into a completely new substance. (Accept any reasonable def

Q2	Exothermic: b) Releases heat | Endothermic: b) Gets cold
Q3	a) EXO b) ENDO c) EXO d) ENDO
Q4	Exo = Exit = Heat Exits = Gets HOT | Endo = Enter = Heat Enters = Gets COLD
Q5 ★	Accept: Burning wood, campfire, sparkler, digestion of food, hand warmers, welding, etc.

■ SECTION 7: HOMEWORK & EXTENSION ACTIVITIES

Science Spy Challenge (All Students)

Find **ONE** chemical reaction at home tonight that involves heat. Record the following:

- **Name of reaction:** ________________________________
- **What gets hot or cold?** _______________________________
- **Is it exothermic or endothermic?** ____________________
- **Signal word that helped you decide:** ___________________

Share with the class the next day! Try to find something no one else will think of. ■■

Extension Activities (Advanced / Gifted Students)

Activity	Description
Research Project	Find out how hand warmers work. Write 3–4 sentences explaining whether they are exothermic or
Creative Writing	Write a short story (10–12 sentences) from the point of view of a heat particle in an exothermic rea
Experiment Design	Design a safe at-home experiment to test whether mixing two kitchen ingredients produces heat o
Chart Making	Draw a colourful chart showing 5 examples of each type of reaction with drawings and labels.

■ SECTION 8: STUDENT QUICK REFERENCE CARD

Print and distribute to students as a take-home reference card. Students can paste it in their science notebooks.

■ EXOTHERMIC	■ ENDOTHERMIC

EXO = EXIT Heat Exits → Gets HOT	ENDO = ENTER Heat Enters → Gets COLD
Signal: hot, flame, burning, releases heat, gives off heat	Signal: cold, absorbs heat, cools down, needs heat
Examples: • Burning diya • Coal in sigri • Cooking dal • Rusting iron • Burning crackers	Examples: • Clay pot water • Ice pack • Photosynthesis • Baking soda + vinegar • Cold packs